



A PRACTICAL PLAYBOOK

Competing on Kaggle

and building cross-validation you can trust

For competitors who know the basics and want to climb the
leaderboard the reliable way.



What separates climbers from plateauers



Generalization, not rank

The best competitors optimize a score that holds on unseen data — not the public board of the day.



Disciplined iteration

Change one thing at a time and measure it against a single, fixed validation scheme.



Diversity, then ensemble

Build several genuinely different models and blend them late — that's where the last gains live.

Anatomy of a competition



The metric is law

Optimize the exact evaluation metric. Convenient proxies (accuracy for an AUC task) quietly cost you rank.



Public vs private leaderboard

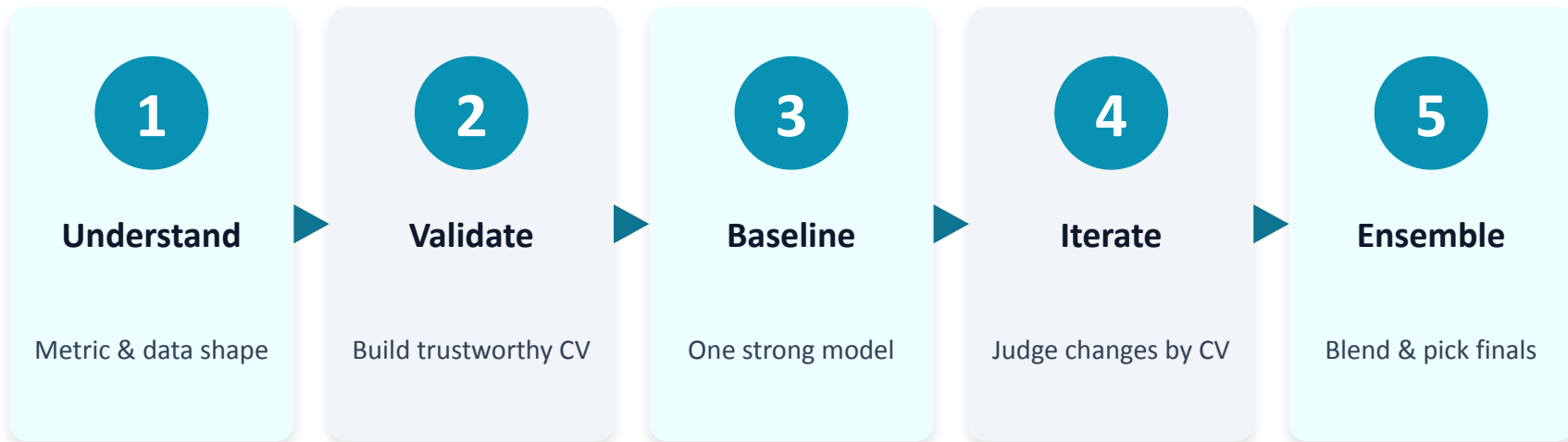
The public board scores a small slice of the test set; the private board, hidden until the end, decides the result.



Mind the shake-up

Competitors who tune to the public slice often tumble when the private scores are revealed. Don't be one of them.

A workflow that compounds



The same loop runs many times — each pass is a CV-checked experiment, not a leaderboard gamble.

The public leaderboard can possibly be a trap

- The public score is computed on a small, fixed subset — sometimes only a few hundred rows.
- Submit enough times and you can 'learn' that subset by luck, not by genuine skill.
- Local cross-validation uses all of your labels, giving a far more stable estimate of true performance.



Public leaderboard

One noisy probe of a tiny, fixed slice — and limited submissions to spend on it.



Local cross-validation

k held-out tests averaged over all your data — repeatable and yours to trust.

How k-fold cross-validation works

- Split the training data into k folds (5 is a common default).
- Train on k-1 folds, validate on the held-out fold; rotate k times.
- Average the k scores → your CV score, plus a sense of its variance.
- **Every row is used for validation exactly once.**

5-fold split



Choose the right CV scheme



Stratified k-fold

Classification, especially imbalanced.
Preserves class ratios in every fold.



Group k-fold

Repeated entities (users, patients). Keep
each group entirely within one fold.



Time-series split

Temporal data. Always train on the past and
validate on the future.



Plain k-fold

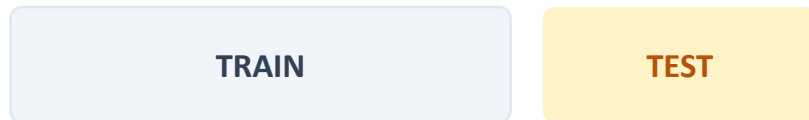
Large, i.i.d. tabular data with no grouping or
time structure to respect.

THE KEY PRINCIPLE

Make your CV mirror the test split

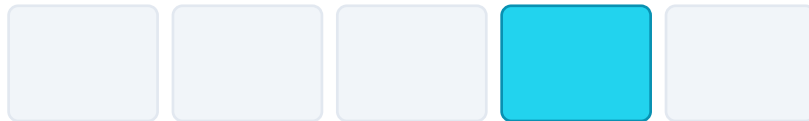
- Ask how the organizers split train vs test — by time? by group? by stratum?
- Reproduce that split locally so CV simulates the real generalization gap.
- If the test set holds unseen users, your folds must hold out whole users too.

Competition



split by X (time / group / stratum)

Your local CV



same split logic, applied to your folds

A CV that matches the split is your single biggest source of leaderboard correlation.

Don't leak the answer into your CV



Leaks (avoid)

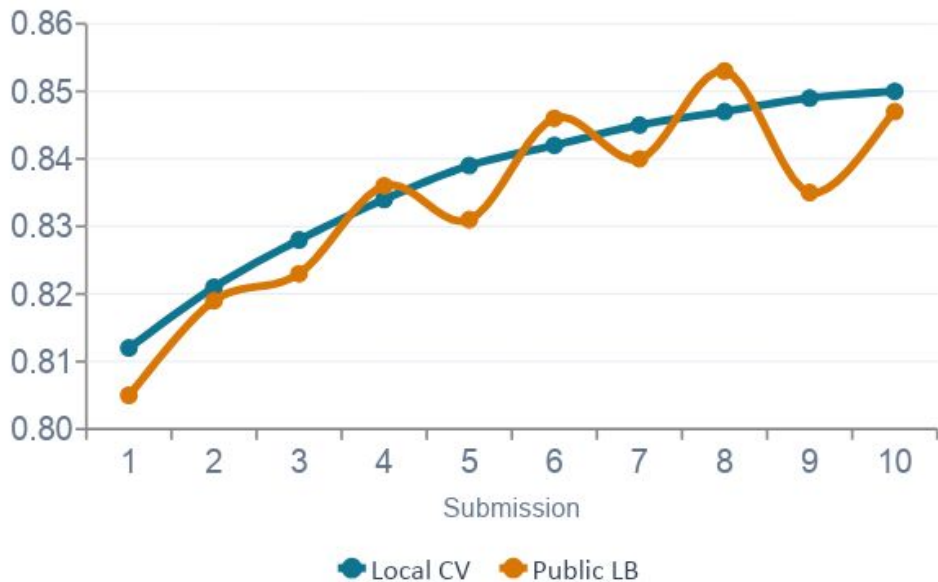
- Scale or encode using full-data statistics before you split.
- Select features by peeking at the whole dataset.
- Let duplicates or group members straddle train and validation.



Clean (do)

- Fit every transform on the training fold only, then apply it.
- Do feature selection inside the CV loop, fold by fold.
- Keep groups intact and deduplicate before splitting.

When CV and LB disagree, trust CV



- Track both every submission and watch how they move together.
- CV up, LB flat is usually still real — the LB sample is just noisy.
- LB up, CV flat often means you're overfitting the public subset.

Select your final submissions by CV — not by public rank.

From validation to ensembling



Save out-of-fold (OOF) predictions

Each fold predicts its held-out part; stitched together they cover every training row, leak-free.



Blend diverse models

Average or weight different model families — GBDTs, neural nets, linear — using OOF scores to set the weights.



Stack carefully

Train a meta-model on OOF predictions. Powerful, but easy to overfit — validate the stack with the same rigour.

Climb with a compass, not a guess



Optimize the exact evaluation metric.



Build a CV that mirrors the train / test split.



Fit every transform inside the fold — no leakage.



Trust CV over the public leaderboard.



Save OOF predictions from day one.



Ensemble diverse models; pick finals by CV.